LaTeX Cheatsheet

A comprehensive reference for academic writing — document structure, math, figures, tables, and bibliography.

DOCUMENT SKELETON

```
\documentclass[11pt,a4paper]{article}
\usepackage[utf8]{inputenc}
\usepackage[T1]{fontenc}
\usepackage{amsmath, amssymb, amsthm}
\usepackage{graphicx}
\usepackage{hyperref}
\usepackage[margin=1in]{geometry}
\usepackage{cite}

\title{Title}
\author{Author \\ Affiliation}
\date{\today}

\begin{document}
\maketitle
\begin{abstract} ... \end{abstract}
\section{Introduction}
\bibliographystyle{plain}
\bibliography{refs}
\end{document}
```

DOCUMENT CLASSES

CLASS	USE FOR
`article`	Papers, short reports
`report`	Theses, multi-chapter docs
`book`	Books with parts
`beamer`	Slide presentations
`letter`	Formal letters
`IEEEtran`	IEEE journal submissions

SECTIONING

`\part{}`	Top-level grouping
`\chapter{}`	Chapter (book/report)
`\section{}`	Section
`\subsection{}`	Subsection
`\subsubsection{}`	Sub-subsection
`\paragraph{}`	Run-in heading
`\section*{}`	Unnumbered variant

TEXT FORMATTING

`\textbf{bold}`	**bold**
`\textit{italic}`	*italic*
`\emph{emphasized}`	context-sensitive emphasis
`\texttt{mono}`	typewriter
`\textsc{Caps}`	small caps
`\underline{}`	underline
`{\large …}`	size: tiny → Huge

MATH MODE

Inline vs Display

`$E = mc^2$` → $E = mc^2$

`\[ x = \frac{-b \pm \sqrt{b^2-4ac}}{2a} \]`

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

SYMBOL	CODE
$\alpha\ \beta\ \gamma\ \delta$	`\alpha \beta \gamma \delta`
$\sum\ \prod\ \int\ \oint$	`\sum \prod \int \oint`
$\partial\ \nabla\ \infty$	`\partial \nabla \infty`
$\leq\ \geq\ =\ \approx$	`\leq \geq \neq \approx`
$\subset\ \in\ \cup\ \cap$	`\subset \in \cup \cap`
$\to\ \mapsto\ \Rightarrow$	`\to \mapsto \Rightarrow`
$\hat{x}\ \bar{x}\ \tilde{x}\ \vec{x}$	`\hat \bar \tilde \vec`
$\mathbb{R}\ \mathcal{L}\ \mathfrak{g}$	`\mathbb \mathcal \mathfrak`

COMMON CONSTRUCTS

`\frac{a}{b}`	$\frac{a}{b}$
`\sqrt[n]{x}`	$\sqrt[n]{x}$
`x^{n}, x_{i}`	x^{n}, x_{i}
`\sum_{i=1}^{n}`	$\sum_{i=1}^{n}$
`\int_{a}^{b}`	$\int_{a}^{b}$
`\binom{n}{k}`	$\binom{n}{k}$
`\lim_{x \to 0}`	$\lim_{x \to 0}$

EQUATION ENVIRONMENTS

```
% Numbered single equation
\begin{equation}
  f(x) = \int_0^x e^{-t^2}\, dt
\end{equation}

% Multi-line aligned at &
\begin{align}
  (a+b)^2 &= a^2 + 2ab + b^2 \\
  (a-b)^2 &= a^2 - 2ab + b^2
\end{align}

% Cases
f(x) = \begin{cases}
  x^2 & x \geq 0 \\
  -x  & x < 0
\end{cases}

% Matrix
\begin{pmatrix} a & b \\ c & d \end{pmatrix}
```

THEOREMS & PROOFS

```
\newtheorem{theorem}{Theorem}
\newtheorem{lemma}[theorem]{Lemma}
\theoremstyle{definition}
\newtheorem{definition}{Definition}

\begin{theorem}[Pythagoras]
  $a^2 + b^2 = c^2$.
\end{theorem}

\begin{proof}
  ...
\end{proof}
```

LISTS

```
\begin{itemize}
  \item Bulleted item
\end{itemize}

\begin{enumerate}
  \item Numbered item
\end{enumerate}

\begin{description}
  \item[Term] definition
\end{description}
```

FIGURES

```
\begin{figure}[!htbp]
  \centering
  \includegraphics[width=0.7\linewidth]{plot.pdf}
  \caption{Caption goes here.}
  \label{fig:plot}
\end{figure}

% Reference: see Figure~\ref{fig:plot}
```

Placement: `h` =here, `t` =top, `b` =bottom, `p` =float page, `!` =override LaTeX preferences.

TABLES

```
\begin{table}[!htbp]
  \centering
  \begin{tabular}{l c r}
    \hline
    Method & Acc. & Time \\
    \hline
    Baseline & 0.82 & 12s \\
    Ours     & \textbf{0.91} & 8s \\
    \hline
  \end{tabular}
  \caption{Results.}
  \label{tab:results}
\end{table}
```

Columns: `l` left, `c` center, `r` right, `p{3cm}` wrapped, `|` vertical rule. Use `booktabs` with `\toprule \midrule \bottomrule` for publication-quality tables.

CITATIONS & BIBLIOGRAPHY

```
% In preamble
\usepackage{natbib}  % or biblatex

% In text
\cite{smith2020}            % [1]
\citep{smith2020}           % (Smith, 2020)
\citet{smith2020}           % Smith (2020)
\cite[p.~42]{smith2020}     % with page

% refs.bib
@article{smith2020,
  author = {Smith, J.},
  title  = {Paper Title},
  journal= {J. of Things},
  year   = {2020},
  volume = {12},
  pages  = {1--20}
}
```

CROSS-REFERENCES

`\label{eq:foo}`	Place anchor
`\ref{eq:foo}`	Number only
`\eqref{eq:foo}`	(n) for equations
`\pageref{}`	Page number
`\autoref{}`	"Figure 3" (hyperref)

Convention: prefix labels by type — `fig:`, `tab:`, `eq:`, `sec:`, `thm:`.

SPACING & BREAKS

`\\` or `\newline`	Line break
`\par` or blank line	New paragraph
`\newpage` / `\clearpage`	Force new page
`\quad \qquad`	Horizontal space
`\vspace{1em}`	Vertical space
`~`	Non-breaking space
`\noindent`	Suppress paragraph indent

SPECIAL CHARACTERS

Reserved: `& % $ # _ { } ~ ^ \` → escape with `\` (or `\textbackslash`, `\textasciitilde`, `\textasciicircum`).

Quotes: ``` ``word'' ``` → "word" · **Dashes:** `-` hyphen, `--` en-dash, `---` em-dash · **Ellipsis:** `\ldots`

HYPERREF & LINKS

```
\usepackage[
  colorlinks=true,
  linkcolor=blue,
  citecolor=blue,
  urlcolor=blue
]{hyperref}

\href{https://trybibby.com}{Bibby AI}
\url{https://trybibby.com}
```

COMPILATION PIPELINE

STANDARD PIPELINE (WITH BIBLIOGRAPHY):

```
pdflatex paper.tex
bibtex paper  (or biber for biblatex)
pdflatex paper.tex  (resolve refs)
pdflatex paper.tex  (finalize)
```

Modern alternative: `latexmk -pdf paper.tex` handles all passes automatically.

QUICK DIAGNOSTICS

ERROR	LIKELY CAUSE
`Undefined control sequence`	Missing `\usepackage` or typo
`Missing $ inserted`	Math symbol used in text mode
`Overfull \hbox`	Long word/URL — use `\sloppy` or `microtype`
`Reference undefined`	Recompile (refs need 2 passes)
`File ended while scanning`	Unclosed `{` or environment